DIT 204: Programming with Python

OPERATORS AND EXPRESSIONS

Learning Objectives

By the end of this lesson, students should be able to:

1. Define operators and expressions.
2. Identify different types of operators in Python.
3. Perform arithmetic calculations using operators.
4. Use comparison operators to make decisions.
5. Apply logical operators in expressions.
6. Understand operator precedence.
7. Solve real-world problems using operators.

Introduction to Operators

An **operator** is a symbol used to perform an operation on one or more operands (values or variables).

Example:

```
a = 10
b = 5

sum = a + b
```

In this example:

- + is the operator
- a and b are operands

What is an Expression?

An **expression** is a combination of values, variables, and operators that produces a result.

Examples:

```
5 + 3
age > 18
marks >= 50
```

Types of Operators in Python

Python provides several categories of operators:

1. Arithmetic Operators
2. Assignment Operators
3. Comparison (Relational) Operators
4. Logical Operators
5. Membership Operators
6. Identity Operators

Arithmetic Operators

Used for mathematical calculations.

Operator	Meaning	Example
+	Addition	5 + 2
-	Subtraction	5 - 2
*	Multiplication	5 * 2
/	Division	5 / 2
//	Floor Division	5 // 2
%	Modulus	5 % 2
**	Exponent	5 ** 2

Examples

Addition

```
a = 10
b = 20

print(a + b)
```

Output:

30

Multiplication

```
length = 5
width = 4

area = length * width

print(area)
```

Output:

20

Modulus Operator (%)

Returns the remainder.

```
print(10 % 3)
```

Output:

1

Application:

Checking whether a number is even or odd.

```
number = 12

print(number % 2)
```

Output:

0

Assignment Operators

Used to assign values to variables.

Operator	Example
=	x = 5
+=	x += 3
-=	x -= 3
*=	x *= 3
/=	x /= 3

Example

```
x = 10

x += 5

print(x)
```

Output:

15

Comparison Operators

Used to compare values.

Result is either:

True

or

False

Operator	Meaning
==	Equal to
!=	Not equal to
>	Greater than
<	Less than
>=	Greater than or equal to
<=	Less than or equal to

Example

```
age = 20

print(age >= 18)
```

Output:

True

Real-Life Application of Comparison Operators

A university requires students to score at least 40 marks to pass.

```
marks = 55

print(marks >= 40)
```

Output:

True

Logical Operators

Logical operators combine conditions.

Operator	**Meaning**
and	Both conditions must be true
or	At least one condition must be true
not	Reverses the result

AND Operator

```
age = 20
citizen = True

print(age >= 18 and citizen)
```

Output:

True

OR Operator

```
print(5 > 10 or 10 > 5)
```

Output:

True

NOT Operator

```
print(not True)
```

Output:

False

Membership Operators

Used to test whether a value exists in a sequence.

Operator	Meaning
in	Exists
not in	Does not exist

Example

```
name = "Python"

print("P" in name)
```

Output:

True

Identity Operators

Used to compare memory locations.

Operator	Meaning
is	Same object
is not	Different object

Example:

```
x = 5
y = 5

print(x is y)
```

Output:

True

Operator Precedence

Python follows mathematical order of operations.

Example:

```
result = 5 + 2 * 3
print(result)
```

Output:

11

First:

2 * 3 = 6

Then:

5 + 6 = 11

5.12 Using Parentheses

Parentheses have highest priority.

```
result = (5 + 2) * 3
print(result)
```

Output:

21

Practical Examples

Example 1: Area of Rectangle

```
length = 10
width = 5

area = length * width

print(area)
```

Example 2: Student Average

```
maths = 80
science = 70
english = 90

average = (maths + science + english) / 3

print(average)
```

Example 3: Even or Odd

```
number = 15

print(number % 2 == 0)
```

Output:

False

Common Errors Students Make

Mistake 1

Using

=

instead of

==

Incorrect:

if age = 18:

Correct:

if age == 18:

Mistake 2

Forgetting brackets.

Incorrect:

average = marks1 + marks2 + marks3 / 3

Correct:

average = (marks1 + marks2 + marks3) / 3

Class Exercises

Exercise 1

Write a program that calculates the area of a circle using:

Area = 3.142 × r × r

Exercise 2

Write a program that accepts two numbers and displays:

- Sum
- Difference
- Product
- Quotient

Exercise 3

Write a program that checks whether a student has passed using:

marks >= 50

Assignment

1. Differentiate between operators and expressions.
2. State five arithmetic operators in Python.
3. Explain the difference between = and ==.
4. Write a program to calculate the perimeter of a rectangle.
5. Write a program to determine whether a number entered by a user is even or odd.

Quiz Questions (Application-Based)

Question 1

A shop owner wants a program that automatically calculates a customer's total bill. Which category of operators would be used most frequently?

Question 2

A university admission system requires applicants to be at least 18 years old. Which operator would be most appropriate for checking this condition?

Question 3

A student enters a number and wants to know whether it is even or odd. Which arithmetic operator would be most useful and why?

Question 4

Why might a programmer use parentheses when calculating a student's average score?

Question 5

A voting system requires a person to be at least 18 years old **and** be a registered citizen before voting. Which logical operator should be used and why?